

Task 2 - Sentinel-2 Image Matching

Potential improvements

1. Dataset Expansion and Diversity

The current dataset covers a single tile (36UYA) across three seasonal dates, which limits generalization to new geographies and land-cover types. To address this, the following should be incorporated:

Multi-tile, multi-biome coverage: 8-15 tiles spanning forest, steppe, urban, mountainous, and coastal terrain, each across four seasons, to reduce overfitting to a single landscape;

L2A (atmospherically corrected) imagery: replacing L1C top-of-atmosphere data removes atmospheric noise as a confound between seasons and provides the SCL layer for principled cloud/land-cover filtering;

Non-planar and radiometric augmentation: local elastic deformations and sensor/atmospheric noise simulation, in addition to the current global homography, since real seasonal change is not a pure planar transform.

2. Training Procedure Fixes

Fine-tuning on a small, narrow dataset surfaced concrete failure modes that should be corrected before scaling up:

BatchNorm stabilization: freezing BatchNorm running statistics when unfreezing the backbone, since small batch sizes (4) caused noisy statistics and measurable quality regression versus the pretrained baseline;

Complete loss function: adding the fine-level (subpixel) regression loss, currently missing, which directly affects match confidence and localization precision;

Progressive unfreezing and early stopping: differential learning rates for backbone vs. matching heads, and validation-based early stopping instead of a fixed epoch count, to avoid catastrophic forgetting on a narrow dataset.

3. Evaluation and Benchmarking

Quantitative evaluation currently relies only on synthetic homography pairs, which does not capture real seasonal appearance change:

Independent ground truth: a held-out set of real cross-season pairs with GT from manual control points or an external co-registration tool (e.g., AROSICS);

Standard matching metrics: Mean Matching Accuracy (MMA) at multiple error thresholds and AUC for homography estimation, rather than a single reprojection-error number;

Baseline comparison and out-of-distribution testing: benchmarking against SIFT+RANSAC, SuperPoint+LightGlue, and vanilla LoFTR, and testing on entirely new tiles/regions, not just a held-out date of the training tile.

4. Full-Scene Processing

The pipeline currently operates on fixed 512x512 patches; production use requires handling full Sentinel-2 scenes (10980x10980 px):

Tiling and stitching: overlapping-patch inference with result merging into a single global coordinate system;

Inference optimization: GPU batching, mixed-precision (FP16/BF16), and optionally ONNX/TensorRT export to meet production latency requirements.